

N SAI VIGHNESH

AI Engineer · Full-Stack Developer

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PROFESSIONAL SUMMARY

Computer Science undergraduate (9.3 CGPA) building AI-powered applications with hands-on experience in LLM integration, prompt engineering, Linux-native AI diagnostics, and deep learning model training and deployment (PyTorch, DCGAN). Built TermBrain, an AI CLI assistant that compresses Linux logs into LLM-ready context windows and provides read-only fix suggestions. Trained generative models and deployed them as live demos on Hugging Face Spaces with Gradio. Builds full-stack interfaces (React, Next.js, Spring Boot) to serve AI-driven features to end users. Oracle Cloud certified in Generative AI and DevOps. National-level hackathon participant (Meta PyTorch OpenEnv, Guidewire DEVTrails).

PROJECTS

TermBrain — Linux-Native AI Diagnostic CLI

Jan 2026 – Present

Python, Typer, Rich, g4f, SQLite, Linux, systemd | [GitHub](#)

- Built a Linux-native AI diagnostic CLI with diagnose, suggest-fix, doctor, security, monitor, history, and schedule command groups for terminal-first debugging.
- Implemented regex-based log compression that extracts error/fail/critical lines and recent context to keep massive terminal logs within LLM context limits.
- Added SQLite-backed history, smart caching, semantic search, Rich live markdown streaming, role-based prompts, and read-only fix suggestions through g4f cloud inference.

GAN-Based Synthetic Image Generator

Jan 2026

Python, PyTorch, DCGAN, Gradio, Hugging Face Spaces | [Live Demo](#) | [GitHub](#)

- Implemented and trained vanilla GAN and DCGAN architectures in PyTorch to generate synthetic images of diseased plant leaves, augmenting limited training datasets for agricultural ML models.
- Compared training stability and output fidelity across multiple generator/discriminator configurations, documenting loss curves and sample quality at each training checkpoint.
- Deployed the generator as an interactive Gradio application on Hugging Face Spaces with CPU-optimized preprocessing for real-time inference.

ClipFlow AI — Developer Productivity Dashboard

Feb 2025

React, Next.js, PostgreSQL, AI/LLM Integration | [Live Demo](#) | [GitHub](#)

- Integrated an AI assistant that analyzes user-submitted algorithms and provides step-by-step hints and optimization suggestions.
- Built and deployed a full-stack productivity platform on Vercel that aggregates coding activity into a unified analytics dashboard with personalized streak tracking.
- Implemented server-side rendering via Next.js with PostgreSQL for persistent user data and session management.

Travel Planner — AI-Assisted Itinerary Builder

Dec 2025

React, Vite, Spring Boot, MySQL, Gemini API | [GitHub](#)

- Integrated the Gemini API to power a conversational chatbot that generates personalized day-wise travel schedules based on user preferences, budget, and travel dates.
- Built the full-stack serving layer with a React frontend and Spring Boot REST API backend, implementing prompt engineering and response parsing for structured AI-generated itinerary output.

TECHNICAL SKILLS

AI / ML: PyTorch, Scikit-learn, g4f, Prompt Engineering, Gemini API

Languages: Python, Java, JavaScript, C, C++, SQL

LLM & Data: SQLite Caching, Semantic Search, Gradio, NumPy, Pandas

Front-End: React.js, Next.js, HTML5, CSS3, Vite

DevOps & MLOps: Docker, Kubernetes, Jenkins, Hugging Face, Git

Back-End: Spring Boot, FastAPI, REST APIs, Supabase

EDUCATION

B.Tech, Computer Science & Engineering — KL University, Guntur, AP

Aug 2023 – May 2027

CGPA: 9.3 / 10

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | [Verify](#)

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | [Verify](#)

Oracle Cloud Infrastructure 2025 Certified DevOps Professional | [Verify](#)

HackerRank Software Engineer Role Certification | [Verify](#)

HACKATHONS & COMPETITIONS

- Meta PyTorch OpenEnv Hackathon x Scaler (Mar 2026)** — Developed robust Reinforcement Learning agent environments using Meta's OpenEnv framework for complex continuous control scenarios.
- Guidewire DEVTrails Hackathon 2026** — Built "GigShield," a 3-tier AI parametric insurance platform for Indian gig workers with ML-powered fraud detection and automated claim triggers using Scikit-learn.
- Design Thinking & Innovation @ KL-CIE (Dec 2023)** — Selected for Mock Shark Tank (Project Expo-3rd Edition) for "E-Waste Management" project, demonstrating design thinking and product pitch skills.